

```
CREATE TABLE Customers (  
    CustomerID NUMBER PRIMARY KEY,  
    Name VARCHAR2(100),  
    Age NUMBER,  
    Balance NUMBER(10,2),  
    IsVIP VARCHAR2(5)  
);
```

```
CREATE TABLE Loans (  
    LoanID NUMBER PRIMARY KEY,  
    CustomerID NUMBER,  
    InterestRate NUMBER(5,2),  
    DueDate DATE,  
    CONSTRAINT fk_customer  
        FOREIGN KEY (CustomerID)  
        REFERENCES Customers(CustomerID)  
);
```

```
INSERT INTO Customers VALUES (101, 'Rahul Sharma', 65, 15000, 'FALSE');
```

```
INSERT INTO Customers VALUES (102, 'Priya Singh', 45, 12000, 'FALSE');
```

```
INSERT INTO Customers VALUES (103, 'Aman Das', 70, 8000, 'FALSE');
```

```
INSERT INTO Customers VALUES (104, 'Neha Roy', 30, 5000, 'FALSE');
```

```
COMMIT;
```

```
INSERT INTO Loans VALUES (201, 101, 8.50, SYSDATE + 15);
```

```
INSERT INTO Loans VALUES (202, 102, 9.00, SYSDATE + 20);
```

```
INSERT INTO Loans VALUES (203, 103, 7.50, SYSDATE + 40);
```

```
INSERT INTO Loans VALUES (204, 104, 8.75, SYSDATE + 10);
```

```
COMMIT;
```

Scenerio 1

o **Question:** Write a PL/SQL block that loops through all customers, checks their age, and if they are above 60, apply a 1% discount to their current loan interest rates.

```
DECLARE
    CURSOR c_customers IS
        select CustomerID
        FROM customers
        where age >60;
BEGIN
    For customer in c_customers LOOP
        UPDATE Loans
        SET InterestRate = InterestRate - 1
        WHERE CustomerID = customer.CustomerID;

        DBMS_OUTPUT.PUT_LINE(customer.CustomerID);
    END LOOP;
    commit;
END;
/
```

OUTPUT

The screenshot displays a SQL IDE interface with the following components:

- Code Editor:** Contains the PL/SQL code for the scenario.
- Statement details:** Shows the executed statement and the execution time: "Executed on: Jun 21, 2026, 7:06:44 PM".
- Query result:** Displays the output of the query, showing the customer IDs (101, 103) and a confirmation message: "PL/SQL procedure successfully completed. Elapsed: 00:00:00.027".

The output shows the following results:

```
SQL> DECLARE
    CURSOR c_customers IS
        select CustomerID
        FROM customers
        where age >60;
    BEGIN
        For customer in c_customers LOOP
            UPDATE Loans
            SET InterestRate = InterestRate - 1
            WHERE CustomerID = customer.CustomerID;
            DBMS_OUTPUT.PUT_LINE(customer.CustomerID);
        END LOOP;
        commit;
    END;
/
```

101
103

PL/SQL procedure successfully completed.
Elapsed: 00:00:00.027

Scenerio 2

oQuestion: Write a PL/SQL block that iterates through all customers and sets a flag IsVIP to TRUE for those with a balance over \$10,000.

```
DECLARE

    CURSOR customer IS
    SELECT CUSTOMERID
    FROM customers
    WHERE Balance > 10000;

BEGIN

    FOR cust IN customer LOOP

        UPDATE Customers
        SET IsVIP = 'TRUE'
        WHERE CustomerID = cust.CustomerID;

        dbms_output.put_line(cust.CustomerID);

    END LOOP;

End;

/
```

OUTPUT

The screenshot displays a SQL IDE interface. The main editor on the left shows a PL/SQL block named 'Student.sql*'. The code is as follows:

```
57
58 DECLARE
59     CURSOR customer IS
60     SELECT CUSTOMERID
61     FROM customers
62     WHERE Balance > 10000;
63 BEGIN
64     FOR cust IN customer LOOP
65     UPDATE Customers
66     SET IsVIP = 'TRUE'
67     WHERE CustomerID = cust.CustomerID;
68
69     dbms_output.put_line(cust.CustomerID);
70     END LOOP;
71 End;
72 /
73
74
```

The right pane is titled 'Statement details' and shows the executed statement:

```
DECLARE
    CURSOR customer IS
    SELECT CUSTOMERID
    FROM customers
    WHERE Balance > 10000;
BEGIN
    FOR cust IN customer LOOP
        UPDATE Customers
        SET IsVIP = 'TRUE'
        WHERE CustomerID = cust.CustomerID;

        dbms_output.put_line(cust.CustomerID);
    END LOOP;
End;
```

Below the statement details, there is a button 'Insert Into Editor'. The 'Result' section shows the output of the execution:

```
101
102

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.015
```

The bottom of the IDE shows the 'Query result' tab with the following output:

```
SQL> DECLARE
      CURSOR customer IS
      SELECT CUSTOMERID
      FROM customers...
Show more...

101
102

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.015
```

Scenerio 3

oQuestion: Write a PL/SQL block that fetches all loans due in the next 30 days and prints a reminder message for each customer.

DECLARE

CURSOR c_loans IS

SELECT c.CustomerID,

c.Name,

l.LoanID,

l.DueDate

FROM Customers c

JOIN Loans l

ON c.CustomerID = l.CustomerID

WHERE l.DueDate BETWEEN SYSDATE AND SYSDATE + 30;

BEGIN

FOR msg IN c_loans LOOP

DBMS_OUTPUT.PUT_LINE('Reminder: Dear ' || msg.Name ||

', your Loan ID ' || msg.LoanID ||

' is due on ' || TO_CHAR(msg.DueDate, 'DD-MON-YYYY'));

END LOOP;

END;

/

OUTPUT

The screenshot shows the SQL Developer interface. The left pane contains the PL/SQL code for the block. The right pane is divided into two sections: 'Statement' and 'Result'.

Statement

Executed on: Jun 21, 2026, 7:30:58 PM

```
DECLARE
  CURSOR c_loans IS
    SELECT c.CustomerID,
           c.Name,
           l.LoanID,
           l.DueDate
    FROM Customers c
    JOIN Loans l
    ON c.CustomerID = l.CustomerID
    WHERE l.DueDate BETWEEN SYSDATE AND SYSDATE + 30;
BEGIN
  FOR msg IN c_loans LOOP
    DBMS_OUTPUT.PUT_LINE('Reminder: Dear ' || msg.Name ||
                          ', your Loan ID ' || msg.LoanID ||
                          ' is due on ' || TO_CHAR(msg.DueDate, 'DD-MON-YYYY'));
  END LOOP;
END;
```

Result

Reminder: Dear Rahul Sharma, your Loan ID 201 is due on 06-JUL-2026
Reminder: Dear Priya Singh, your Loan ID 202 is due on 11-JUL-2026
Reminder: Dear Neha Roy, your Loan ID 204 is due on 01-JUL-2026

PL/SQL procedure successfully completed.
Elapsed: 00:00:00.021